

Matthew J. Nitzken

Principal Machine Learning Engineer

Remote - Louisville, KY

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About

I'm a machine learning engineer with more than 15 years in AI. I'm currently a Principal Machine Learning Engineer at DataRobot, where I lead the engineering and architecture for our agentic AI platform and the LLM agent products built on it. I moved onto generative AI in late 2024 and it has been the near total focus of my work since. That work runs across the whole stack: agent architecture, RAG, memory, prompting, tracing, evaluation, and getting all of it into production. Before that I spent my career on full-stack machine learning and data science in large-scale data environments, mostly time series, deep learning, and model explainability. My background is in electrical, computer, and bioengineering, and along the way I've worked in healthcare, data engineering, medical diagnostics and research, image and signal processing, and controls automation. What I gravitate toward hasn't changed. I like hard problems that nobody has solved yet, and I like getting them into production. I enjoy collaborating with, learning from, and mentoring the people around me.

Skills

Generative AI	LLMs & Agents	RAG & Evaluation	Python
Machine Learning	Deep Learning	Time Series	Model Explainability
Cloud & Kubernetes	Data Engineering	Test-Driven Development	Mentorship & Leadership

Education

University of Louisville

Doctor of Philosophy (Ph.D.), in Electrical and Computer Engineering

[Louisville, Kentucky](#)

2010 – 2015

- Dissertation: Shape Analysis of the Human Brain

University of Louisville

Masters of Engineering (M.Eng.), in Bioengineering

[Louisville, Kentucky](#)

2009 – 2010

University of Louisville

Bachelor of Science (B.S.), in Bioengineering

[Louisville, Kentucky](#)

2005 – 2009

Experience

DataRobot

Data Scientist → Senior → Staff → Principal Machine Learning Engineer

[Louisville, KY \(Remote\)](#)

Jul 2019 – Present

Engineering lead and architect for DataRobot's agentic AI platform and the generative AI products built on it. Started as a data scientist and worked through time series, model explainability, and platform architecture before moving to generative AI.

- Moved onto generative AI in late 2024 and built DataRobot's first agent prototypes. I built the deployment automation around them, packaged that as a development template, and kept extending the template until it became the platform our teams and customers build agents on. It supports CrewAI, LangGraph, LangChain, Pydantic AI, and NVIDIA NAT, and carries its own frontends, backends, tracing, evaluation, and prompt tooling. I wrote a large share of it and led the engineers who built the rest.
- Drove the platform's developer experience with UX testing. I sat in on the sessions, found the points where engineers got stuck starting a project, and reworked the templates, guided tasks, and documentation around those points.
- Co-architected af-components, the component framework the platform runs on. Templates drop into a project, bring their own Terraform or Pulumi infrastructure, and stitch together so a full solution deploys in one command. Fixes for bugs, CVEs, and features live in one place and push out to internal and external users as optional updates. It is now open source and became DataRobot's most widely used component library.

- Built more than a dozen end-to-end agentic solutions on the platform for DataRobot's largest accounts, and all of them went to production. NVIDIA and Dell build on the platform, and NVIDIA chose it for their own agentic AI. The work expanded those contracts.
- Applied the same agent tooling to CVE remediation. I built CLIs and agents that patch dependencies, containers, and templates across a large set of repositories, using uv, Jinja, and automated Docker image rebuilds. A team of more than 30 people had been spending roughly 40% of its time on CVE work.
- Joined the Universal SHAP effort while it was in flight and helped carry it to delivery. SHAP doesn't work on time series data natively, so I developed a new method to make it work and ran the analysis to confirm the math held up. I then led the design and engineering to ship it, which put SHAP explanations on more than 200 model blueprints that had none, across DataRobot SaaS and on-prem.
- Reworked a stalled plan for breaking up a core platform monolith into one that could be implemented. The work covered test infrastructure, splitting code into new repositories with their own CI/CD, and reworking core code. Other teams picked up the same approach for their own services.
- Created and led several time series products earlier on. Incremental learning brought online training to the platform and removed a hard ceiling of roughly 10GB on training data. Segmented Modeling split large intermixed datasets into segments and optimized a model for each. I also built DataRobot's early time series trust and explainability features, including feature importance, feature impact, and drift.
- *Experience focused on Python, LLMs, CrewAI, LangGraph, LangChain, Pydantic AI, RAG, evaluation and tracing, Docker, Kubernetes, Helm, Terraform, AWS, GCP, TensorFlow, and TDD*

Humana

Louisville, KY / Boston, MA

Analytics Consultant → Lead Data Scientist

Oct 2016 – Jul 2019

Built and led machine learning across Humana's clinical and Medicare STARS analytics, moving from analytics consultant to lead data scientist across three teams.

- Replaced Humana's Medicare STARS rating forecasts, which were unreliable ARIMA projections, with Monte Carlo simulations (one per plan and measure, plus a global model). They gave a probability curve across the 1 to 5 STARS scale, so leadership could see where each plan was likely to land and its odds of slipping.
- Built a large set of models predicting member health trajectories across conditions common to Medicare members (diabetes, COPD, flu vaccination), ranging from straightforward ML to SAS/Spark-fronted pipelines to TensorFlow deep learning.
- Later joined a new Boston team building real-time deep learning for Medicare members. I migrated data off on-prem to GCP and built the data pipelines and the front-end applications backed by those models.
- *Experience using Python, TensorFlow, Keras, SkLearn, SAS, GCP, AWS, Hadoop, SQL, Airflow, and TDD*

Alliant Technologies

Louisville, KY

Software Engineer → Lead Software and Controls Engineer

May 2015 – Oct 2016

Software and controls developer for automated high-speed sorter systems, promoted to lead the software development and commissioning teams.

- Led software development and controls commissioning through build-out and QA on large-scale material-handling projects.
- Built an Ethernet/IP communications stack for high-speed sortation, and a Python server that simulated the control server so new systems could be tested before deployment.

NeuroAtlas

Louisville, KY

Founder and Machine Learning Engineer

Mar 2013 – May 2015

Co-founder and lead machine learning engineer at NeuroAtlas, building neurological diagnostic software for autism spectrum disorders.

- Went through two healthcare accelerators (XLerateHealth, LaunchIT) and ran several grant-funded initiatives; co-led direction, design, and fundraising.

University of Louisville

Louisville, KY

Graduate Researcher, Bioluminescence Laboratory

Jul 2010 – May 2015

Ph.D. research developing machine learning for digital diagnostics.

- Developed and published 2D/3D shape, image-, and signal-processing algorithms and computer-aided diagnostic ML, across autism, dyslexia, cancer, Alzheimer's, and cardiac/EMG/ECG signals.
- Presented research at national and international conferences, through talks and posters.

Publications and Patents

Please see publications.mattnitzken.com for an up-to-date list of my publications and patents.